Adaptive Trial Design for the RAS-Inhibitor Era

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Today's argument

New RAS therapies need new kinds of trials.

Methods to optimize combination doses are established. Designs that adapt to emerging resistance are enrolling.

The difficulty: getting them calibrated and launched.

The argument of the talk. The science of RAS-era combinations puts real demands on trial design — combination dose-finding under non-monotonic dose-response, and resistance tracking with in-trial adaptation. Modern Bayesian adaptive designs answer both, at different maturity levels: dose-optimization methods are established, resistance-tracking designs are enrolling. What's been missing is the infrastructure to make them practical, and that's where BATON closes the gap. BATON is the closing tool, not the headline. Note the deliberate asymmetry in the fragment — "established" vs "enrolling" — this prevents the resistance-side overclaim that would land badly with a SPORE audience who knows EVOLVE hasn't yet reported.

The clinical case

Why combinations, and why now?

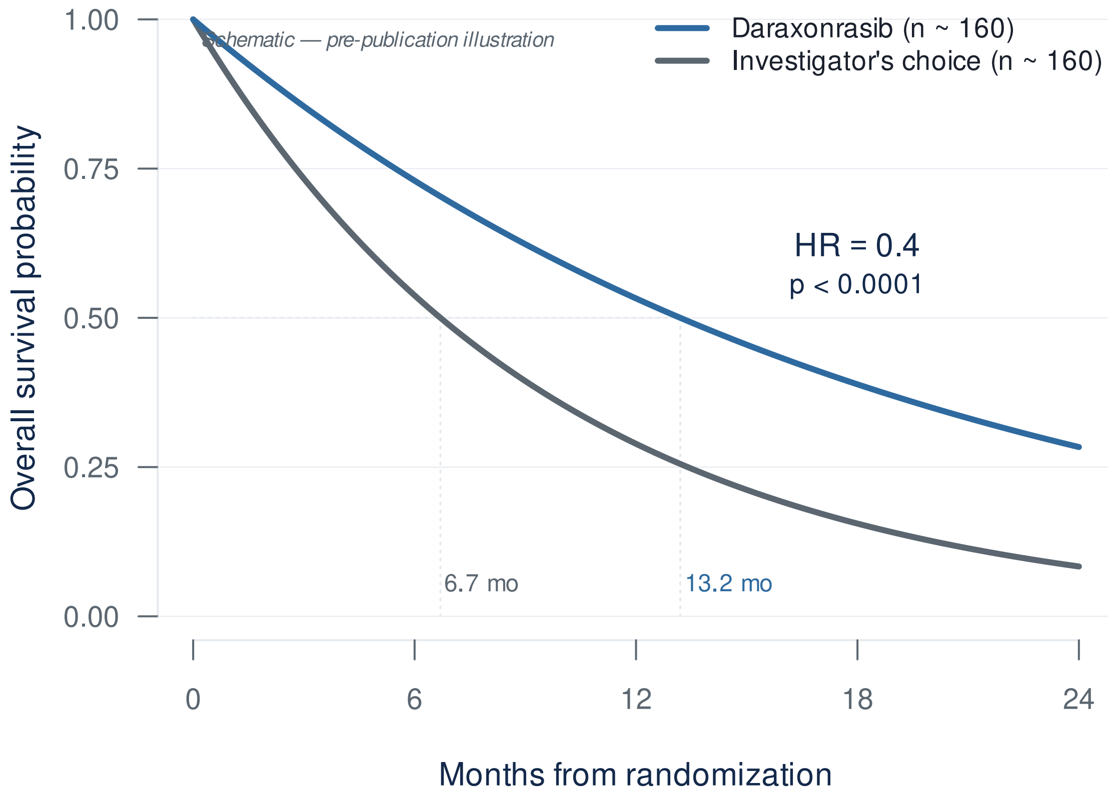
Three slides on the clinical pressure before we get to design problems.

The RAS revolution has arrived for PDAC

- **RASolute 302** (2L mPDAC, *NEJM* 2026)[1]:
  - **Median OS: 13.2 vs 6.7 months**
  - **HR 0.40 — 60% reduction in risk of death** ($p < 0.0001$)
- Daraxonrasib (RMC-6236) — oral, multi-selective RAS(ON) inhibitor

- ASCO 2026 Plenary, May 31 (Wolpin)

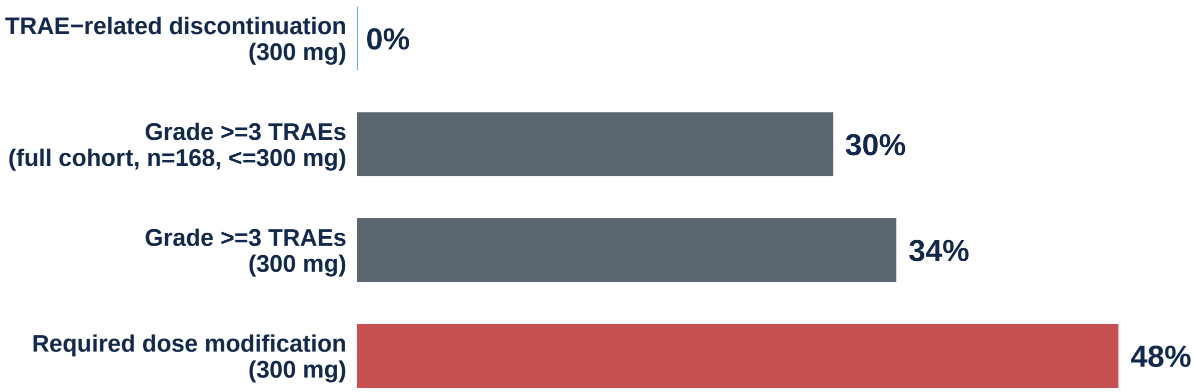
O'Reilly et al., *NEJM* 2026 (RASolute 302) · ASCO Plenary Abstr LBA5



For pancreatic cancer this is the largest survival signal in a generation. Final analysis was presented at the ASCO Plenary on May 31 and published in *NEJM* the same day. Cite the NEJM paper, not a press release — this room will care about that. Two numbers to have in your pocket for Q&A. Median follow-up was 8.5 months at the February 10 data cutoff, so if anyone presses on maturity that's your number. And the 13.2 vs 6.7 figure is overall ITT; the dual primary endpoint was OS and PFS in the RAS G12-mutant subgroup (228 vs 231 patients), where HR was also 0.40 (P = 5.9e-10). If a discussant draws the ITT-vs-primary distinction, the answer is that the 0.40 benefit held in both — robustness across RAS mutation status is the whole point of a multi-selective RAS(ON) inhibitor. That robustness is a strength to lean on, not a vulnerability. Whatever else you take from this talk, this result will reshape the GI oncology trial landscape over the next two to three years — every program in this room will soon be deciding what to combine with this drug, in which biomarker-defined patients.

…but monotherapy has visible ceilings

RMC-6236-001 (PDAC cohort, *NEJM* 2026)[2]



Half of patients on monotherapy at 300 mg already need dose adjustments. Combinations layer on top — with less room to spare.

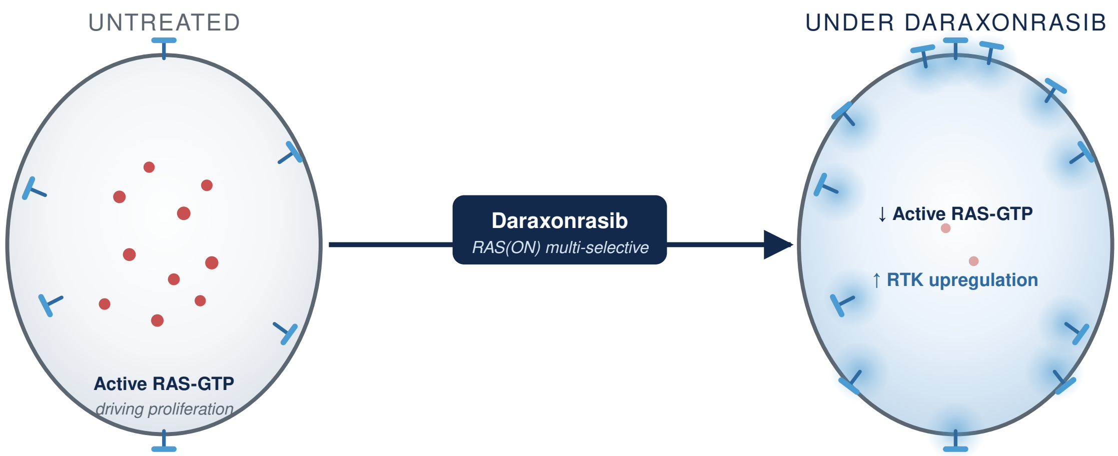
Wolpin et al., *NEJM* 2026 (RMC-6236-001 PDAC cohort)

Three numbers worth keeping in mind. Across the full cohort of 168 patients on doses up to 300 mg, about a third have Grade 3 or worse toxicity. No patient discontinued because of drug toxicity — that's genuinely remarkable for an active drug in this disease. The qualifier matters; everyone with metastatic PDAC eventually comes off for progression, but nobody had to come off this drug for tolerability. Now narrow to the 300 mg dose specifically, which is the Phase 3 monotherapy dose: Grade 3 toxicity is closer to 34%, and forty-eight percent of patients required dose modifications. That 48% is the load-bearing number for the rest of this talk. For a drug being positioned as the backbone of future combinations, that's the headroom you start spending the moment you add a partner. The combination already moved to 200 mg — informally acknowledging the same headroom problem without searching for the actual optimum, which is the next slide. The question isn't "is daraxonrasib tolerable as monotherapy?" — clearly it is. The question is "at what dose, on what schedule, with what partners, in which patients?" That's a trial-design question, and the rest of this talk is about answering it.

Biomarker-driven combinations are the next move

- Daraxonrasib drives **RTK upregulation** on PDAC cell surface — the mechanistic case for biomarker-driven, resistance-preemptive combinations[3]
- **1L combination cohort** (daraxonrasib **200 mg** + GnP, n=40)[4]:

– ORR: **58%** · DCR: **90%** · 6-mo PFS: **84%** · 6-mo OS: **90%**



RTK upregulation is simultaneously a resistance mechanism and a rational combination target

Zhuang et al., *Cancer Res* 2026 (AACR RAS) · Wolpin et al., AACR 2026 LB407

Mallika Singh's group at Revolution Medicines has published the mechanistic case for what comes next. Inhibiting active RAS upregulates receptor tyrosine kinases on the cell surface — which is simultaneously a resistance mechanism and a rational combination target. Their explicit framing is that combinations should be biomarker-driven and resistance-preemptive. This is the manufacturer staking out the next decade of the program. And the first combination cohort has already delivered — 58% ORR with gemcitabine/nab-paclitaxel in first-line metastatic PDAC, 90% disease control, 84% six-month PFS. The combinations are coming. Many of them.

The dose-finding problem

Why MTD only may not be best for RAS combinations, and what replaces it.

Why combinations break the rules you grew up on

- **MTD-only dose-finding was built for single-agent cytotoxics**[5] — 3+3 is the canonical example; CRM, BLRM, mTPI, BOIN all share the paradigm. Assumptions:
  – Monotonic dose-toxicity

  - Cycle-1-observable DLTs
  - Single dose dimension

- **No joint-surface model.** Combinations strain all three assumptions; the structural failure is the missing model.
- **In practice:** pick A's dose, fix it, add B — the dominant pattern. Rivière's review of 162 combination trials: **88% used 3+3 designs** even when both drugs were escalated[6,7].

Le Tourneau et al., *JNCI* 2009 · Rivière et al., *Ann Oncol* 2015

3+3 answered the question it was built for — the MTD of a cytotoxic under a monotonic toxicity assumption. For RAS combinations none of those assumptions hold. The deeper problem is structural and operational. In practice almost no one searches the joint dose surface. The protocol fixes agent A's dose and adds agent B, so two monotherapy designs run in series and the joint surface is never characterized. Thall, Garrett-Mayer, Wages, Halabi, and Cheung made the formal version of this argument in their 2024 *Clinical Trials* paper, the SCT response to Project Optimus: conventional Phase I designs cannot reliably identify safe and effective doses. Q&A POCKET — frequentist alternatives. If a methodologist asks "what about non-Bayesian combination designs?" the honest answer is that very few practical, deployed frequentist combination designs exist. The field has converged on Bayesian / model-assisted designs (POCRM, BOIN-COMB, BLRM, PIPE, surface-free) because combinations require borrowing strength across dose pairs and partial-order constraints — naturally handled by priors and posterior updating, awkwardly handled without them. The cleanest frequentist alternative in the literature is Conaway, Dunbar, & Peddada 2004 (Biometrics 60:661-669), order-restricted isotonic regression for combinations — theoretically clean but not widely deployed. 3+3 generalizations to combinations exist but inherit all the original problems plus combinatorial explosion. Sequential lead-in (find A then add B) is the dominant industry practice — that is what we are criticizing. Q&A POCKET — Rivière backup. The Rivière et al 2015 systematic review of 162 combination phase I trials gives the "almost never" claim a documented count: 88% used 3+3, all but one used a design developed for single-agent evaluation. If anyone challenges "almost never" as an unsupported universal, that is your number. Next slide makes the abstract failure concrete with the Amgen sotorasib + pembrolizumab case.

The Amgen case: sotorasib + pembrolizumab

Same sponsor, two different designs:

- **Monotherapy (CodeBreaK 100):** Bayesian model-based, single-agent surface
- **Combination:** IO dose fixed, only sotorasib varied — joint surface never modeled

The defining toxicity:

- Severe **immune-mediated** hepatotoxicity, not predictable from the single agent
- **88%** of Grade 3-4 events outside the Cycle 1 DLT window[8]

What MTD-only dose-finding was never built to handle, all at once: joint surface unsearched, non-additive toxicity, late-onset toxicity.

Hong et al., *NEJM* 2020 (CodeBreaK 100) · Li et al., *JTO* 2022 (WCLC OA03.06)

This is the worked example of the principle on the previous slide. The Amgen case is the clean one — and notice it is not a competence problem. Same sponsor used a model-based Bayesian logistic regression for sotorasib monotherapy in CodeBreaK 100. But for the sotorasib + pembrolizumab combination in CodeBreaK 100/101, they fixed the IO dose, explored only sotorasib across cohorts, and never modeled the two-drug surface. Even the most sophisticated sponsors collapse the combination to a one-drug search — which is the gap BATON closes. The combination's defining toxicity proved the point twice over: severe hepatotoxicity you could not predict from the single agent, and 88% of the Grade 3-4 events landed outside the 21-day DLT window — so a Cycle 1 design would have missed most of it. Q&A POCKET — where 88% comes from. The 22 of 25 numerator and denominator is the audit-ready figure. Source: Li et al WCLC 2022 OA03.06 / JTO 2022 supplement. If pressed, this single example demonstrates everything 3+3 was never built to handle, all at once: joint surface unsearched (combination not modeled — 3+3 has no joint-surface model), toxicity non-additive (synergistic hepatotoxicity), and late-onset toxicity outside Cycle 1 (88% out-of-window — 3+3 assumes Cycle-1-observable DLTs). Note: don't claim this violates monotonic dose-toxicity — they didn't search the surface, so we can't show non-monotonicity here. Q&A POCKET — "Why didn't they use a late-onset-aware design?" TITE-BOIN (Yuan 2018) and TITE-CRM (Cheung 2000) handle late-onset toxicity by extending the assessment window beyond Cycle 1 and allowing dose decisions while earlier cohorts' toxicity data are still pending. The methods have existed for over five years. But no published RAS Phase I trial has adopted them: sotorasib (CodeBreaK 100) used BLRM with 21-day cycles; adagrasib (KRYSTAL-1) used mTPI with standard Cycle-1; daraxonrasib (RMC-6236) uses standard Bayesian dose-escalation with 21-day cycles. This is the design-framework layer of the deck's deployment-gap thesis: the tool exists, the empirical case for needing it is well-established, adoption is the gap. Q&A POCKET — mechanism. The severe hepatotox is hypothesized to be immune-mediated, not pharmacologic. Sotorasib's pro-inflammatory effect on the tumor microenvironment can re-activate the latent T-cell response that pembro left in place — biopsies in patients with severe sotorasib + ICI hepatotox show patterns consistent with ICI-induced hepatitis. Sequencing matters: in a real-world series, 58% Grade 3 hepatotox occurred when sotorasib started within 30 days of prior ICI. Same immune mechanism explains both "unpredictable from single agent" and "late-onset" — ICI-mediated toxicity has weeks-to-months kinetics, not days. Sources: JTO 2023 brief report (Begum et al), JCO PO 2024 case series (Adagrasib after sotorasib-related hepatotox).

Why MTD-only dose-finding doesn't fit targeted agents and immunotherapies

- **Pembrolizumab** — efficacy plateaus at the lowest dose tested; MTD never reached[9]
- **CAR-T** — non-monotonic; high doses drive severe CRS/ICANS, not benefit

- **A RAS combination spans both shapes.** Neither rewards titrating to toxicity.

The reason MTD fails for these agents is mechanistic, and it has two flavors. I'll illustrate with pembrolizumab and CAR-T — non-RAS examples, but the cleanest in the literature, and the same dose-response shapes show up in the IO and bispecific partners daraxonrasib is going to be combined with. The first flavor you all know from pembrolizumab — the efficacy curve plateaus very early. In the original pembrolizumab studies, response rates across 2 mg/kg, 10 mg/kg every three weeks, and 10 mg/kg every two weeks were statistically indistinguishable. The exposure-response curve was flat. The MTD was never reached because efficacy never improved with dose. The 200 mg flat dose used today was selected by population PK modeling to reproduce the exposures seen at 2 mg/kg in a typical patient — a weight-based-to-flat conversion of the lowest active dose, not a search for the optimum. The second flavor is the umbrella. CAR-T is the canonical case. Very high cell doses unambiguously drive severe CRS and ICANS without proportional benefit. The downslope itself is product-specific — some show a true decline, others a plateau or threshold — but what is universal across CAR-T and bispecifics is the absence of a monotonic increase, and that alone is enough to break titrate-to-toxicity. These are not RAS drugs, but they are the cleanest pictures of the two shapes, and they are the shapes your RAS combinations will actually have. Daraxonrasib is the plateau — its monotherapy program never cleanly reached an MTD, had no DLTs, and 300 mg was selected on tolerability, which is exactly why 300 versus 200 is still a live question. The partners you'll add bring the umbrella. A RAS combination lives across both panels, and that is the whole reason MTD logic cannot find its optimal dose. You need a design that watches both dimensions.

The cost of MTD-only dose-finding

- **Sotorasib** (KRAS G12C, first-in-class, approved 2021)[10]:
  - 960 mg selected as the **highest dose tested**, no DLTs — a more-is-better default, not an MTD
  - FDA required a **240 mg vs 960 mg** comparison. **PFS was similar** at one-quarter the exposure[11]
- **MTD optimal dose** for targeted agents and immunotherapies
- **More than 40% of patients** in small-molecule registrational trials require dose reductions or interruptions[12]

Same drug class as daraxonrasib.

Hong et al., *NEJM* 2020 (CodeBreaK 100) · Hochmair et al., *Eur J Cancer* 2024 · FDA–AACR, *Clin Cancer Res* 2025

The empirical evidence that this is not just theoretical. The FDA-AACR Strategies for Optimizing Dosages series, published in Clinical Cancer Research in 2025, reports that more than

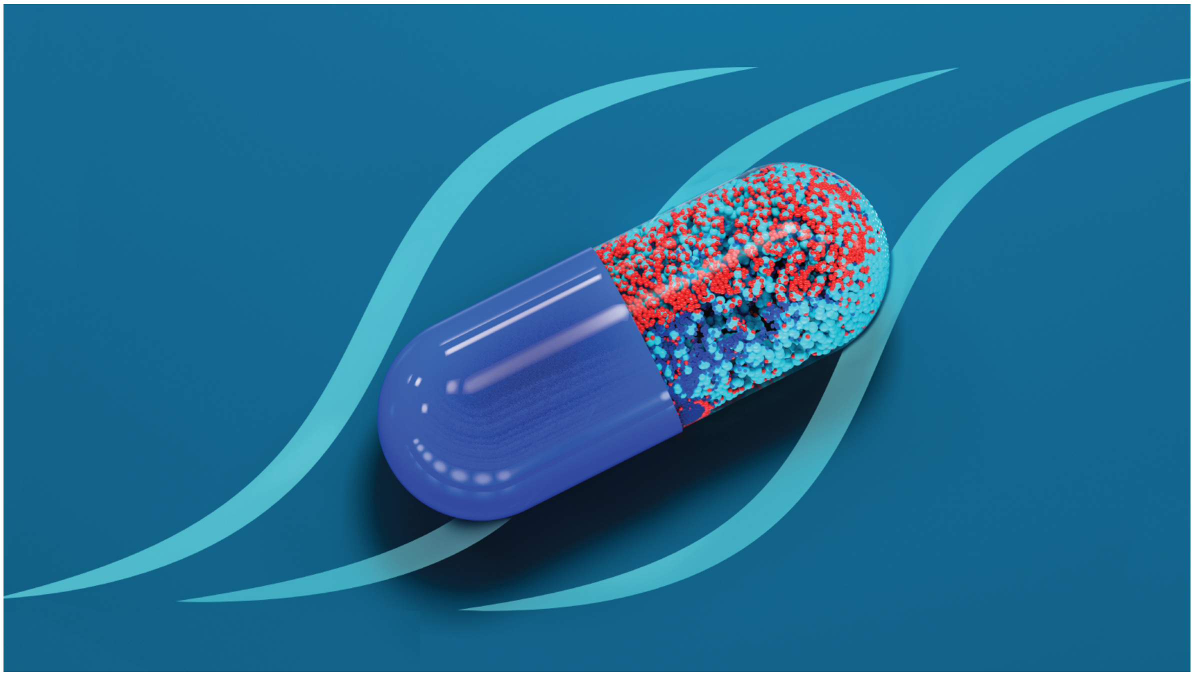
40% of patients in small-molecule registrational trials require dose reductions or interruptions due to overly aggressive dosing — and links that pattern to potentially accelerated emergence of drug resistance, which quietly foreshadows the back half of this talk. The case you all know best is sotorasib, the first-in-class KRAS G12C inhibitor, approved in 2021. The Phase I program never reached an MTD because there were no DLTs; 960 mg was simply the highest dose tested, a more-is-better default rather than an MTD-driven choice. The FDA later required a post-marketing comparison against 240 mg, a fourfold lower dose. PFS at 240 mg was similar to PFS at 960 mg — the clean evidence that the approved dose was four times higher than needed on the primary endpoint. Note: I am deliberately not folding in the October 2023 ODAC vote on CodeBreaK 200 here. That 10-2 vote was about whether the confirmatory PFS read-out could be reliably interpreted given trial-conduct issues — informative censoring, imbalanced early dropout, investigator-assessed imaging — not about the dose. Conflating it with the dose story is the move a regulatory-savvy attendee will flag. The dose story alone is airtight; keep it that way. For daraxonrasib — same drug class — getting this wrong in Phase I means every combination on top of it inherits an unnecessarily toxic starting point.

Project Optimus: the regulatory ground has shifted

- FDA final guidance, 2024[13]
- Sponsors must **study more than one dose** and justify on efficacy and safety
- **Maximum Tolerated Dose → Optimal Biological Dose (OBD)** — a dose optimized on benefit-risk

OBD = the dose that **optimizes the efficacy–toxicity trade-off** — chosen by weighing the value of a response against the cost of toxicity, not by escalating until DLTs cap the dose[14,15].

Thall et al. *Clin Trials* 2024: conventional Phase I designs cannot reliably identify safe and effective doses[16].



FDA Project Optimus · Oncology Center of Excellence

FDA Guidance for Industry, 2024 (Project Optimus) · Thall et al., *Clin Trials* 2024

What you may not have caught is that the FDA has formalized this — and it lands directly on the MTD problems we've just walked through (shapes don't fit MTD logic; the sotorasib 960-vs-240 case shows the cost). The 2024 Project Optimus guidance moves oncology dose-finding from "maximum tolerated dose" to a dose optimized on benefit-risk — what the field calls the OBD. The guidance language is "optimized dosage," and the field's OBD shorthand maps to it directly. If you submit a Phase I protocol today that only characterizes the MTD, your reviewers are going to push back. This is no longer a methodological preference held by biostatisticians; it's the FDA's stated preference. Now, on what the guidance actually expects: the FDA's stated default is to take two doses into Phase II — the MTD and a dose below it — and use a randomized comparison to determine which is better on benefit-risk. For a single agent that works. It does not scale to combinations, because you cannot randomize across a two-dimensional joint dose grid. That is exactly where model-based joint designs stop being a preference and start being the only practical way to meet the benefit-risk standard, which is where the next slide goes.

How OBD selection actually works

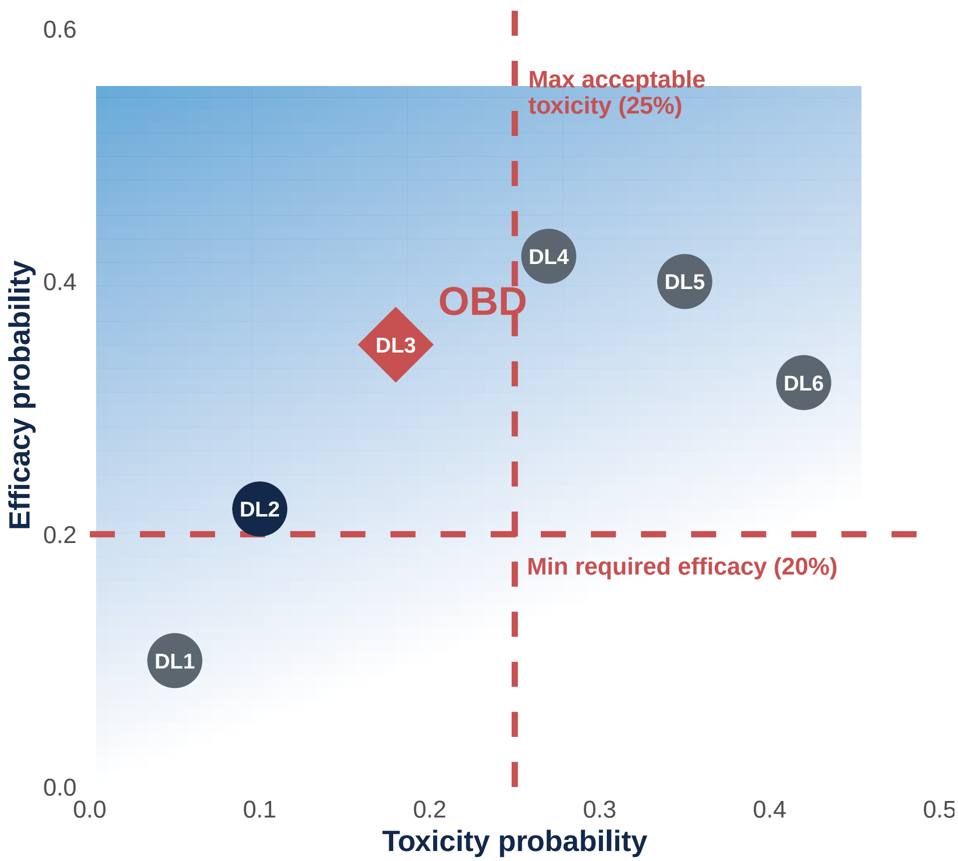
Utility-based dose-finding, in one sentence.

"A response is worth this; a Grade 3 toxicity costs this."

The team writes that utility down up front. The design — Bayesian, model-assisted — searches inside the safe envelope for the dose that **maximizes benefit-risk**, not toxicity.

These methods predate Optimus by 5–15 years. Optimus made what they deliver the regulatory standard[13].

Implementations: **BOIN family** for printed decision tables[14,17]; **POCRM, BLRM** for combinations[18,19]; **TITE-BOIN** for late-onset toxicity[20].



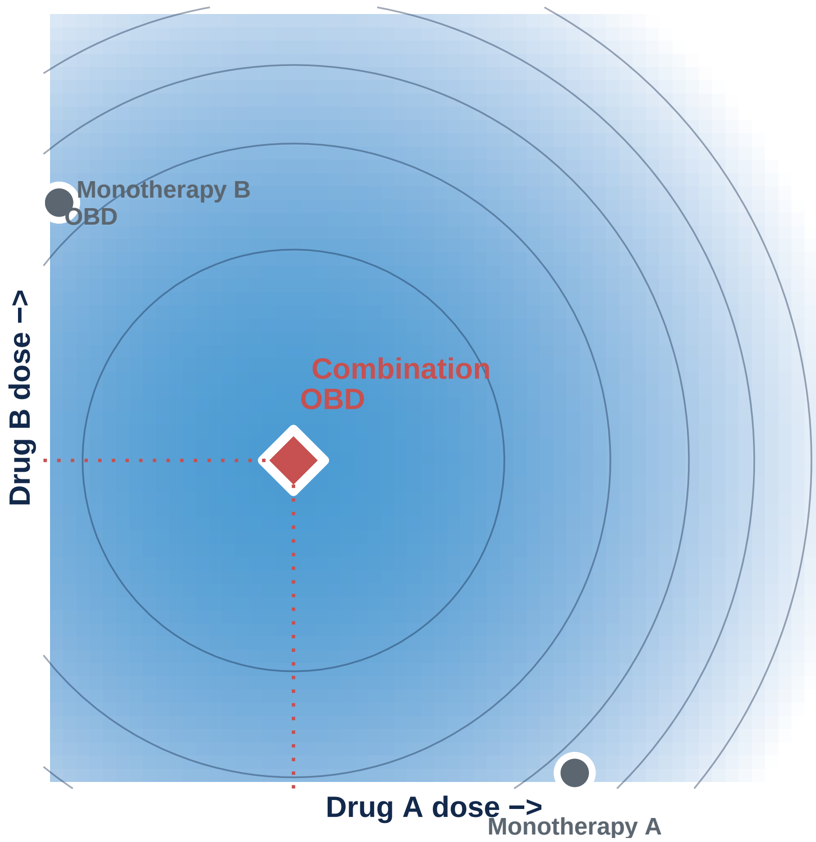
Each dose-level plots on the toxicity × efficacy plane. Admissible doses (inside the red dashed lines) are evaluated for utility; the **OBD** is the admissible dose with the highest utility.

Liu & Yuan 2015 · Zhou, Lee & Yuan 2019 (U-BOIN) · FDA Optimus Guidance

The how, in one sentence: before the trial starts, the team writes down a utility — how do you weigh a response against a Grade 3 toxicity? — and the design searches inside the safe toxicity envelope for the dose that maximizes that utility. (Optimus already established the WHAT on the previous regulatory slide; this slide answers HOW.) The utility becomes the navigation system. The design uses Bayesian / model-assisted rules to keep the trial inside a safe toxicity envelope while searching for the dose that maximizes the agreed-upon benefit-risk utility. What comes out is not a maximum tolerated dose — it's an OBD with a characterized benefit-risk profile. That's the dose Project Optimus is asking sponsors to deliver. PREDATES OPTIMUS — important to land verbally: U-BOIN was published in 2019, BOIN in 2015, TITE-BOIN in 2018, POCRM in 2011, BLRM in 2008. Optimus is 2021/2024. So Optimus didn't create these designs — it codified what they were already doing. The regulatory shift validated 5-15 years of methodological work. ACRONYM POCKETS for Q&A — keep them OFF the visible slide. The BOIN family is the workhorse implementation: BOIN itself for single-agent (Liu and Yuan 2015), BOIN-COMB for drug combinations, TITE-BOIN for late-onset toxicity, BOIN12 for utility-based phase I/II, U-BOIN for the explicit two-stage utility design. POCRM (Wages and Conaway) and BLRM (Neuenschwander, Branson, Gsponer) are the established combination alternatives. mTPI is another common model-assisted variant; PIPE is a dual-agent product-of-independent-Beta-probabilities design (Mander 2015) for completeness. If a methodologist asks "which one?" the answer is: in the U.S. and academic groups, BOIN/U-BOIN dominate for the printed-decision-table operational simplicity; Novartis uses BLRM by default; the rest are specialty tools picked for specific constraints. Credibility line — leading RAS Phase I programs already run on Bayesian designs: sotorasib used BLRM in CodeBreaK 100, adagrasib used mTPI in KRYSTAL-1. RAS Phase I uniformly uses Cycle-1 windows; none have adopted TITE-variants despite the sotorasib + pembro late-tox finding — that's a Q&A pocket from slide 9. For this audience the one-sentence takeaway is: modern Phase I optimizes benefit-risk instead of maximizing toxicity. Don't recite the acronym list out loud; only deploy it if a methodologist asks. Next slide extends this from single-agent to combinations.

Combination dose selection in RAS is harder

Combinations make MTD-only dose-finding worse on every axis.



Conceptual schematic of the benefit-risk surface. The combination OBD sits below both single-agent OBDs; single-agent data alone cannot reconstruct the contour.

- The combination OBD can sit **below both single-agent OBDs**
- Single-agent data **cannot reconstruct** the joint surface
- For RAS, the **toxicity margin is already spent** at the monotherapy dose

Fix what you know. Search what you don't.

RMC-GI-102 (daraxonrasib + GnP) fixed daraxonrasib at 200 mg without searching the joint surface[4].

Hong et al., *NEJM* 2020 · Wolpin et al., *NEJM* 2026 (RMC-6236-001)[2,10]

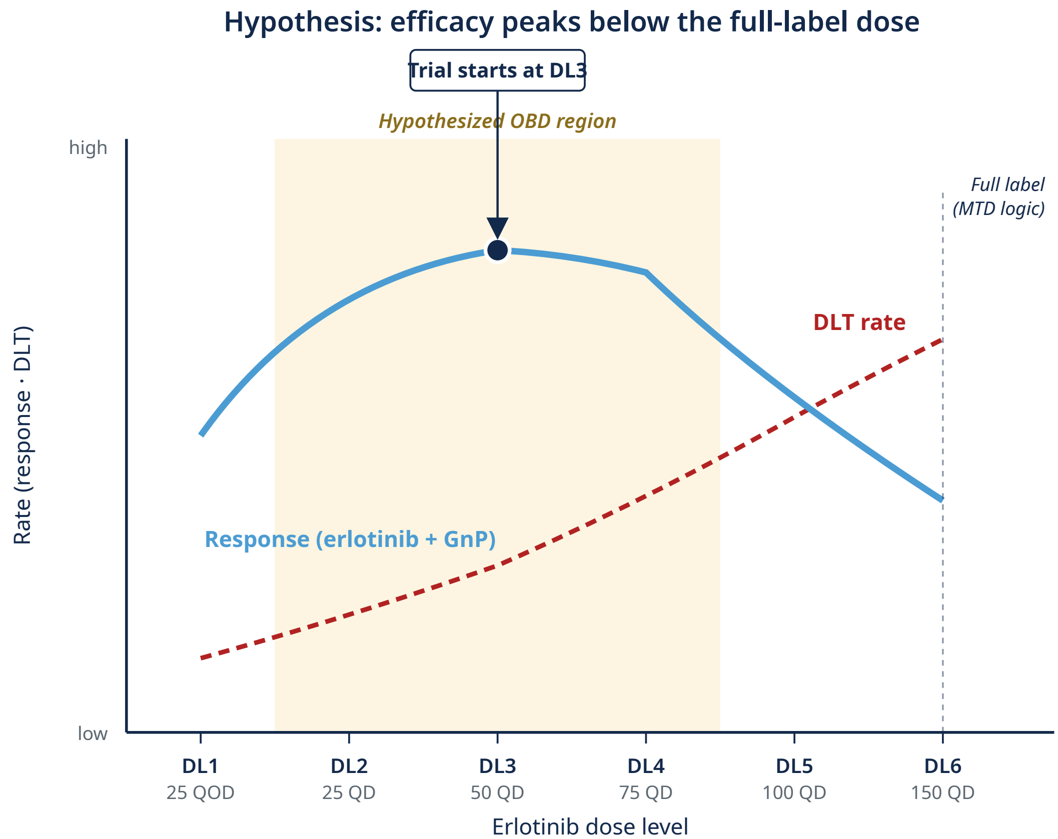
The previous slide showed HOW OBD selection works for a single agent. This slide extends it to combinations. The job of this slide is to name what changes when you go from one drug to two. Three things to land verbally. First, you are no longer choosing one dose, you are choosing a joint pair (A, B), and the single-agent OBDs do not reconstruct the joint surface — sotorasib's 960-versus-240 result tells you the surface looks different from what the single-dose escalation suggested. Second, effects interact on both efficacy and toxicity, so the joint OBD can sit BELOW either monotherapy OBD on each axis (shown by the dotted projections in the schematic). Third, and this is the RAS-specific part: the monotherapy toxicity is already substantial — 34% G 3 at 300 mg, 48% dose modification — so you start the combination with less margin to spend. The first combination cohort already moved daraxonrasib down to 200 mg with GnP, but we do not know whether 200 mg is right or whether 150 mg or some other dose is right given other partners. The "fix what you know, search what you don't" callout is the operational rule: reduce to one drug only when the fixed component's dose is externally established — spend the search on the drug whose combination-optimal dose is genuinely uncertain. SCHEMATIC DISCIPLINE: the chart shows efficacy contours only as a conceptual proxy for benefit-risk. The actual OBD is on a benefit-risk surface (efficacy AND toxicity, per slide 13). Don't claim the chart shows the literal joint efficacy-toxicity utility; it's a one-dimensional reduction for visual purposes. Next slide: PANGEA as the applied trial.

PANGEA: U-BOIN in basal-like PDAC

- **Trial:** LCCC2220-DCT (Somasundaram, PI), basal-like PDAC by **PurIST**[21]
- **Combination:** erlotinib + biweekly gemcitabine/nab-paclitaxel
- **Hypothesis:** lower-dose erlotinib may be more efficacious (OBD-below-MTD)[22]
- **Six dose levels:** 25 mg QOD → 25 / 50 / 75 / 100 / 150 mg QD; **start at DL3**

Design: *Stage 1 BOIN*[17] (target DLT 25%, cohorts of 4) → *Stage 2 BOIN12-like utility within U-BOIN*[14] (tox 25%, eff 20%). **Up to 52 patients; ~28 at OBD** (Scenario 1).

Rashid et al., *Clin Cancer Res* 2020 (PurIST) · Cohen et al., *Cancer Chemo Pharm* 2016 · Liu & Yuan 2015 · Zhou, Lee & Yuan 2019 (U-BOIN)

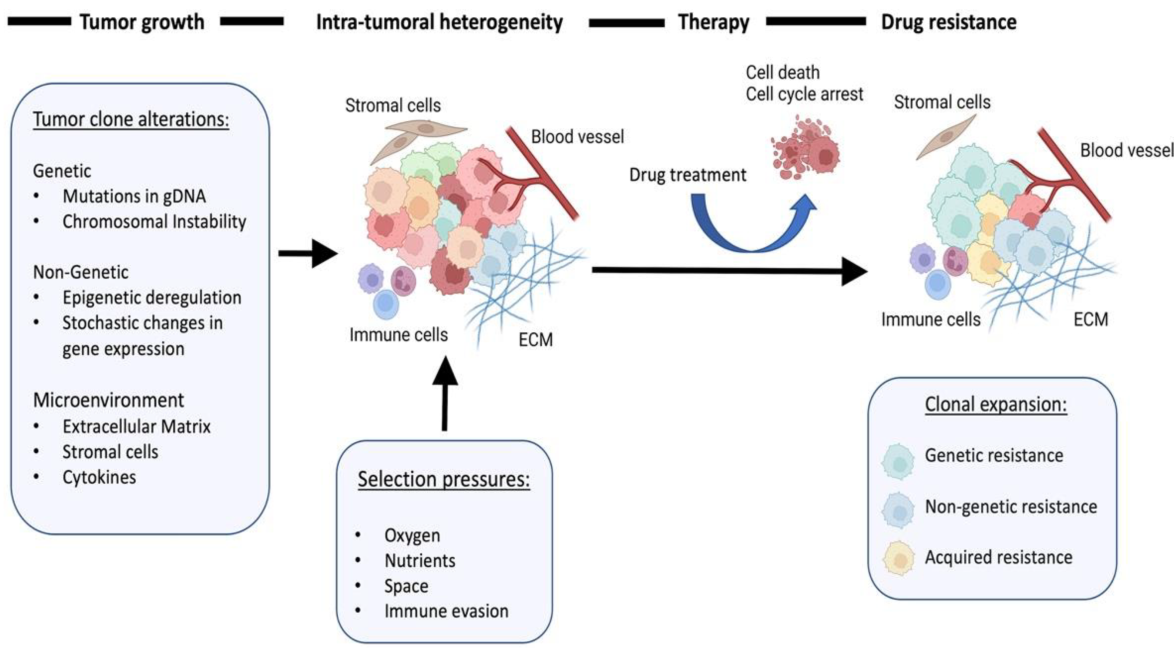


Here's the design in a real PDAC trial at UNC. PANGEA, led by Ashwin Somasundaram. The population is biomarker-defined — basal-like PDAC by PurIST, a subtype that has historically responded poorly to gemcitabine/nab-paclitaxel. The intervention is adding low-dose erlotinib to that backbone. And here's where the talk's argument lands on a real protocol: the central biological hypothesis of PANGEA is that lower doses of erlotinib may be more efficacious than the labeled dose, when given in combination with chemotherapy. The full-label dose of erlotinib is 150 mg daily. PANGEA tests six dose levels going down to 25 mg every other day. This is the OBD-below-MTD case made concrete — exactly the kind of question 3+3 cannot answer, and exactly the kind of question U-BOIN was built for. Stage 1 uses BOIN-style toxicity rules to establish the safe envelope, with a target DLT rate of 25%. Once we have enough data at any dose, we transition to Stage 2, which uses a utility function — elicited from our clinical team — to find the dose that maximizes the benefit-risk tradeoff. Doses have to clear both toxicity admissibility, no more than 25% DLT, and efficacy admissibility, at least 20% response rate. The trial ends when twenty patients have accumulated at any one dose, or when fifty-two patients have been treated total. Under the most plausible scenario from our simulations, we expect about thirty-eight patients in the U-BOIN phase, with about twenty-eight ultimately treated at the OBD. Q&A POCKET — justified dimensionality reduction. If a methodologist asks "why are you only searching erlotinib, not the joint surface?": GnP is fixed because the FDA-approved standard-of-care dose is established (MPACT); erlotinib is searched because its combination OBD-below-MTD is the genuinely open question. This is a justified single-axis search inside a two-drug regimen, not a regression to the fix-A-add-B antipattern. This is what an OBD-paradigm Phase I trial looks like in basal-like PDAC.

Resistance is the next problem

Phase 2 onward: *each patient is a moving target*

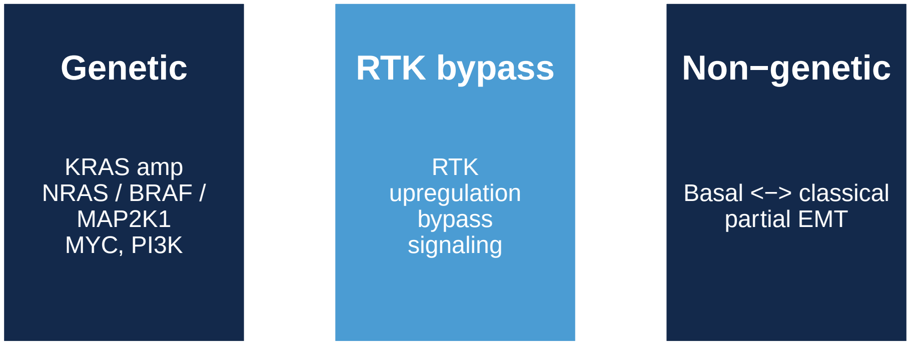
Resistance is a dynamic process



Schematic adapted from ARPA-H ADAPT program overview materials.

Resistance mechanisms emerge under RAS-inhibitor pressure

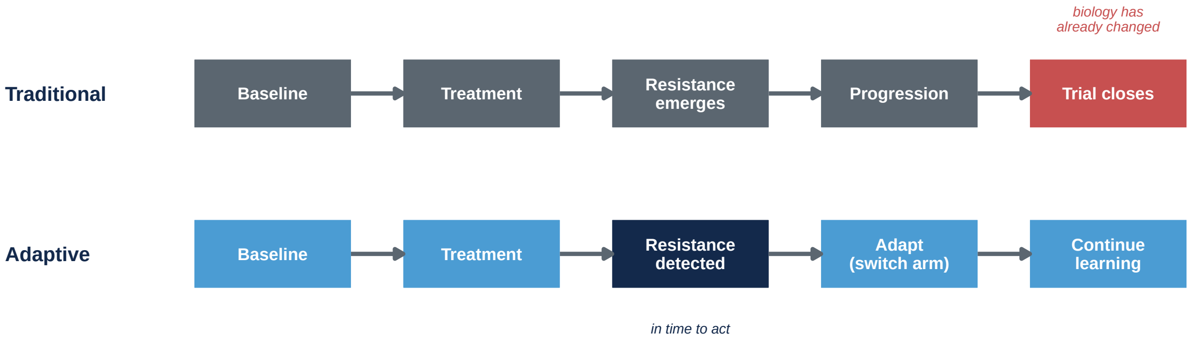
Daraxonrasib monotherapy in mPDAC (n=44, paired ctDNA): **59% show RAS-pathway resistance at progression — KRAS amplification in 36%**[23]



[3,24–27]

Classical trial design treats patients as if they're draws from a fixed distribution — enroll, randomize, administer the regimen, measure outcomes at predefined timepoints. That logic works when the underlying biology is static. Under sustained RAS-inhibitor pressure it doesn't. The tumor evolves. Each patient is a moving target. Concretely — and this is the headline I want to anchor on — Aronchik's ENA 2025 paired-ctDNA work on the drug we just saw on slide 4, daraxonrasib, shows that 59% of PDAC patients on monotherapy develop detectable RAS-pathway resistance alterations at progression, with KRAS amplification specifically in 36%. Same drug from RASolute 302, same disease, n=44 patients with paired pre- and end-of-treatment ctDNA. That is the dynamic biology the trial design has to be prepared to see. Resistance signals start appearing weeks into therapy — sometimes earlier. The mechanisms split into a few buckets. Pathway reactivation through KRAS amplification or downstream NRAS, BRAF, MAP2K1 mutations — Aguirre's PDAC clinical work and Awad's adagrasib resistance series. Downstream amplification — MYC and PI3K family — is Wasko's preclinical RMC-7977 finding (the prevalent in-vivo resistance mechanism in KPC tumours, not pathway reactivation per se since pERK remained suppressed in resistant tumours). RTK upregulation and bypass signaling — the Zhuang and Aronchik mechanistic case for combinations. Cell-state plasticity between basal-like and classical with partial EMT — Aguirre again, plus the Yeh and Johnson labs' kinome work at UNC. None of these are baseline phenotypes. The biology is genuinely dynamic — and that's the setup for what trials need to do differently, which is the next slide.

Why classical trial designs miss resistance



Genomic → ctDNA · non-genomic / plasticity → biopsies, transcriptomics, imaging. A resistance-aware trial has to **do both** during enrollment[24,25].

Awad et al., *NEJM* 2021 · Aguirre et al., *Cancer Discov* 2024

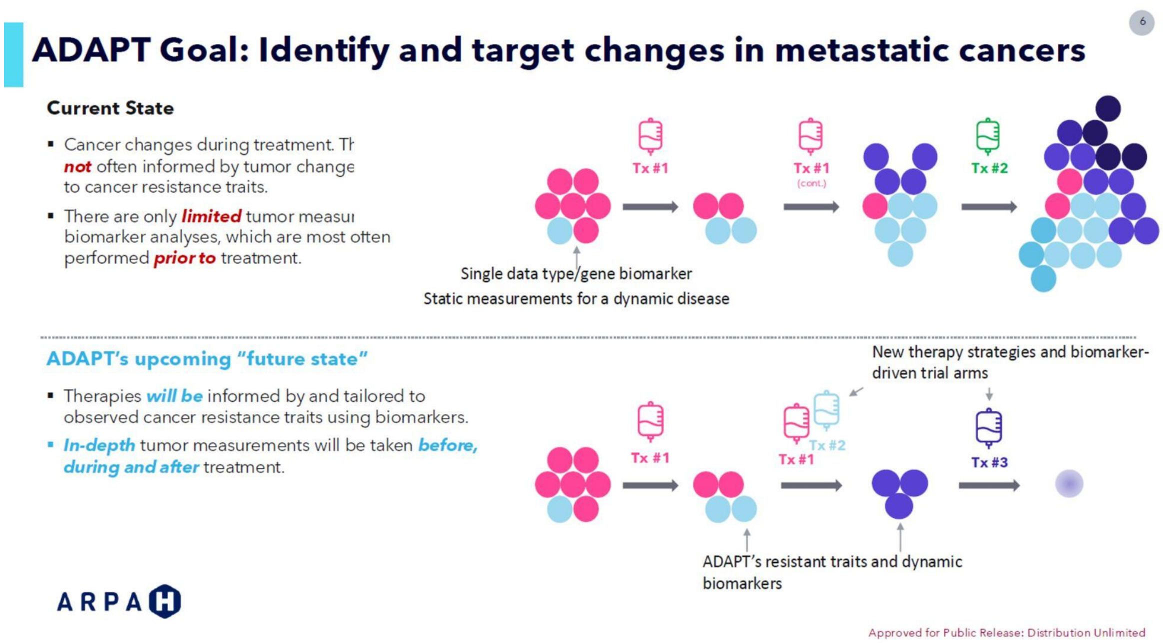
This is the design-implications side of the previous slide. The two-row timeline names the structural failure of classical trial designs under RAS-era biology. The traditional row: baseline → treatment → resistance emerges → progression → trial closes — and the red marker on "trial closes" is the point: by the time the trial reads out, the biology has already changed. The adaptive row catches resistance when it emerges, in time to act on it within the same protocol. Three things to land verbally. First, fixed regimens miss the interception window — by the time the trial reads out, the biology has changed. Second, baseline-only biomarkers can't track evolution under therapy — you measured the wrong thing at the wrong time. Third, resistance is characterized retrospectively, only after progression and trial closure. The Awad and Aguirre papers cited here are themselves post-progression analyses — ctDNA and biopsies collected after patients had already failed therapy. The field's best resistance knowledge was generated exactly the way I'm criticizing — after the fact, outside any prospective design built to catch it. Fourth, the temporal surface — when resistance emerges, in which patients, by which mechanism — is never characterized. The empirical anchor for this room: Aronchik's ENA 2025 paired-ctDNA work on daraxonrasib[23] shows RAS-pathway alterations — KRAS amplification (36% of patients), BRAF, MAP2K1 — emerging at progression in 59% of PDAC patients on monotherapy, mechanisms that a baseline-biomarker trial would not catch. The design implication splits with the biology: genomic mechanisms are catchable by ctDNA — cheap, continuous, easy to add. Non-genomic plasticity needs serial biopsies, transcriptomics, sometimes imaging — expensive, episodic, hard. A resistance-aware trial has to do both, during enrollment, not after. Same structural failure as the combination dose-finding critique two sections ago: static designs for a dynamic problem. Just as the dose problem needed Bayesian designs that handle the joint surface, the resistance problem needs designs that learn during enrollment. Next slide shows what those look like.

ADAPT / EVOLVE: what a resistance-aware trial looks like

ARPA-H's resistance-tracking template.

- Longitudinal resistance detection
- Biomarker-guided adaptation
- Adaptive arm-swaps on emerging biology

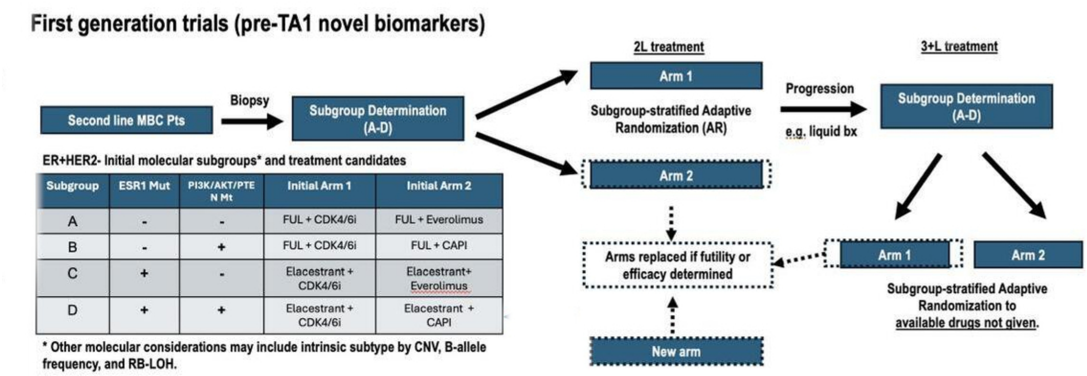
ARPA-H **ADAPT** → **EVOLVE** (UNC-led; Carey PI, Rashid co-PI). **Enrolling.**



ARPA-H ADAPT — current state (top) vs dynamic-biomarker state (bottom).

This is what a resistance-aware trial actually looks like operationally. ARPA-H — the new Advanced Research Projects Agency for Health — has funded a program called ADAPT specifically to build trials of this kind. The ARPA-H figure tells the story in two panels. Top: current state. A single static biomarker, sequential treatments — Tx 1, then Tx 1 continued, then Tx 2 — and the resistant subclones persist into the final cell mix. Bottom: ADAPT's future state. In-depth dynamic biomarkers, treatments switched in response to observed resistance traits, eventual clearance. Our EVOLVE trial in metastatic breast cancer is one of the first to operationalize this, and the design template generalizes directly to the kind of RAS-backbone combination trial you all are going to be running for PDAC in the next few years. CRITICAL FRAMING DISCIPLINE: the closer says "enrolling," not "EVOLVE shows" or "EVOLVE works." This is the resistance-side honesty I committed to in the opening — methods exist and are deployed, not yet completed. If a trialist asks for the readout, the answer is the design exists and is accruing; the readout is what we will report when interim looks are run. Do not let the live delivery drift toward an implied result. The next slide takes the audience inside the protocol — what EVOLVE actually does, and what the PDAC analog would look like.

EVOLVE-BDT — and what this would look like in PDAC



TBCRC EVOLVE-BDT flow — biomarker-stratified subtrials, adaptive randomization, in-trial arm replacement on futility or efficacy.

EVOLVE-BDT (real, enrolling)

ER+/HER2− and TNBC, biomarker-stratified subtrials with Bayesian arm-swap on futility/efficacy.

PDAC analog (hypothetical)

Basal-like vs classical by PurIST[21], same Bayesian arm-swap logic on daraxonrasib-combination arms.

Same architecture, different disease.

This is the load-bearing slide for "could we actually do this in PDAC?" The figure is EVOLVE-BDT's flow — TBCRC EVOLVE-BDT, our group's adaptive platform in 2L metastatic breast cancer. Walk it left to right. Second line MBC patient enters. Biopsy. Subgroup determination — molecular profile assigns A through D. Subgroup-stratified adaptive randomization sends the patient to 2L Arm 1 or Arm 2. Patient is treated. At progression, a liquid biopsy re-determines the subgroup — because the biology has changed — and the patient enters 3+L treatment in a subgroup-stratified Arm 1 or Arm 2. Crucially, the dashed boxes mean arms get replaced when the Bayesian decision rule fires futility or efficacy. New arms enter. The trial keeps learning. Now the columns below the figure are the explicit translation. Left: real EVOLVE specs. ER+/HER2- and TNBC parallel platforms, about 700 patients total. Subgroups A through D defined by ESR1 mutations, PI3K/AKT/PTEN status, PD-L1, androgen receptor — all things that already drive treatment decisions in breast clinical practice. Serial tumor and blood at every visit. Arms swap on the Bayesian decision rule. Right: the PDAC translation. Stratify by basal-like versus classical using PurIST. Subgroups by KRAS amplification, RTK upregulation, partial EMT — the Aronchik data shows 59% develop measurable resistance, so these subgroups have empirical basis. Serial ctDNA for the genomic side, serial biopsies for plasticity — the do-both requirement made concrete. Arms include daraxonrasib plus EGFR-inhibitor for RTK-upregulation patients, daraxonrasib plus SHP2-inhibitor for pathway-reactivation patients, classical-state shifters for plasticity-driven failures. The bottom callout makes the point: same architecture. Every row of the right column has a real biomarker, a real assay, and a real combination partner already in or entering the clinic. This is not a hypothetical that needs new science. It's the deployment problem. Which is exactly what BATON solves. Pause briefly before moving on to the statistical-demands slide.

What this demands of the statistical design

Four design axes any such trial needs.

We have the methods. Deployment is the active work — now.

Four axes that operationalize the resistance-aware trial template. I am showing them to GI SPORE leaders because they are disease-agnostic — requirements for any resistance-aware trial, not properties of breast biology. A RAS-backbone PDAC combination would need all four for the same reasons. Adaptive stopping rules — Bayesian futility and efficacy boundaries that update at every interim look. Real-time biomarker discovery — pre-specified candidate signatures plus an agnostic discovery window so you're not locked into your initial hypothesis. Longitudinal monitoring — serial ctDNA, imaging, biopsies on every patient. This is the "do both" requirement from the mechanisms slide made concrete: ctDNA for the genomic side, biopsies and transcriptomics for the non-genomic plasticity side, on every patient throughout the trial. And in-trial evaluation — biomarker-guided cohorts opened within the same protocol, so when you find a resistance signature you can test the strategy that addresses it during the trial, not as a follow-on study three years later. Now the load-bearing point that sets up the rest of the talk. Continual learning during enrollment is not a tweak to a Phase II — it is a structurally different kind of trial. Each axis adds interacting parameters to the design: interim look timing, posterior-probability thresholds for stopping, biomarker decision rules, randomization-weight updating, cohort-opening logic. Bayesian adaptive designs that handle all of this exist — posterior-probability monitoring, simulation-based operating characteristics, the whole toolkit. What does NOT yet exist routinely is the infrastructure to make those designs deployable — to calibrate them against the operating-characteristic targets that the trial team and the FDA agree on. That's the gap. The next section is about closing it.

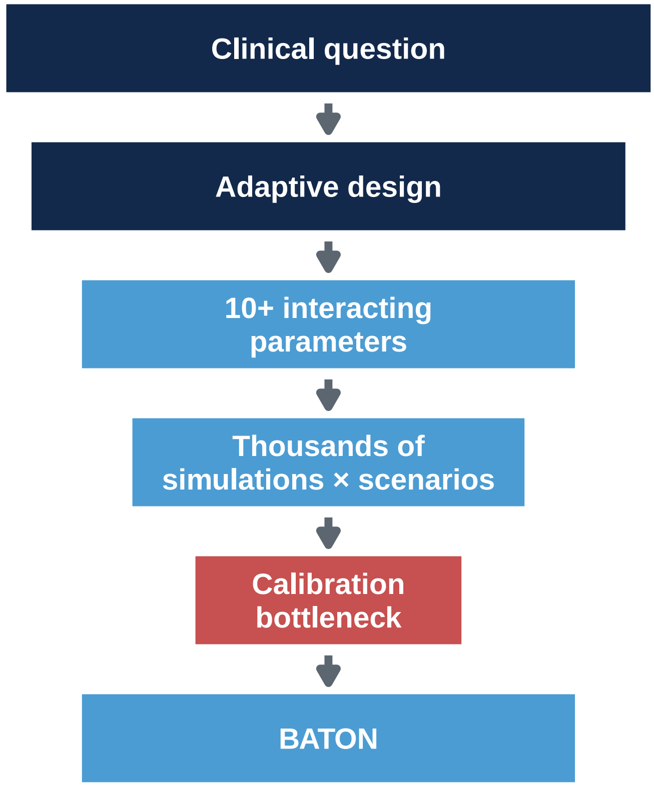
Putting these into practice

Why don't we see more of these trials?

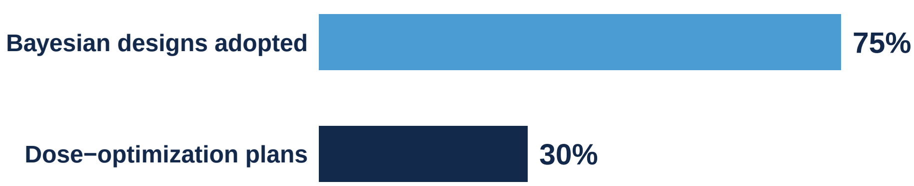
Why don't we see more of these trials?

Funding, accrual, and biopsy logistics are real.

But the barrier that quietly kills good designs — and the one we can actually remove — is calibration.



- **No closed-form for adaptive designs.** Unlike Simon two-stage — where Type I, power, and N come from a formula — adaptive Bayesian designs need *thousands of simulated trials per candidate*: pick parameters (efficacy/futility thresholds, max N, interim timing, H / H assumptions), simulate, read off OCs. **Comparing candidates means simulating each.**
- Design space: **8–13 dimensions** for trials we've calibrated; **20+** for combined dose + resistance. Grid search is infeasible.



Methods adopted. Dose-optimization plans — where calibration cost bites — not[28].

Our group: several months per design. For combined dose + resistance designs, hand-tuning is infeasible.

FDA–AACR Project Optimus impact review, *JCO Onc Adv* 2025 (367 Phase I protocols)

This is the slide that opens the BATON section, and I want to land it carefully. Open with the concession explicitly: funding, accrual, biopsy logistics — these are real, and this room has lived them. I am not standing up here telling you the barrier to better trials is statistics, because that would be a biostatistician naming his own contribution as the field's central obstacle, and you would be right to push back. What I am claiming is narrower and more specific: among the barriers, calibration is the one that quietly kills good designs and the one we can actually remove. You have all watched PIs default back to 3+3 even when they knew a better design existed — the calibration cost was the hidden reason, and that is the gap BATON closes. Now the body. The methods I just walked through — modern dose-finding for combinations, longitudinal biomarker-aware adaptive designs — exist. The reason every clinical trial isn't using them is calibration cost. Every modern design has knobs: target DLT and efficacy rates, utility weights, cohort sizes, stopping rules, sample-size caps. Operating characteristics depend on every knob, in interaction. To know whether a design meets Type I, power, expected N, expected duration, OBD-selection accuracy, and biomarker-promotion error control, you simulate thousands of trial replicates per candidate. And you do this across plausible scenarios. The design space is eight to thirteen dimensions for the trials we've calibrated, and twenty-plus when you combine dose-finding with resistance tracking. A grid search is hopeless — both computationally infeasible and statistically wasteful, because most of the parameter space gives bad designs. In my group, calibrating a single modern combination Phase I/II design by hand takes a senior statistician several months of focused work. For trials that handle dose-finding AND resistance tracking simultaneously, you're at twenty-plus parameters with even more constraints. Hand-tuning doesn't just get slower; it gets infeasible. CLINICAL CONTRAST TO LAND VERBALLY: Most clinicians in the room are familiar with Simon two-stage and similar frequentist designs where you can write down Type I, power, and sample size from a closed-form calculation — plug in the assumptions and out comes the design. For adaptive Bayesian designs there is no such formula. You pick a candidate design — a specific combination of efficacy thresholds, maximum sample size, cohort size, futility/efficacy stopping rules, null and alternative assumptions — then run thousands of simulated trials at that configuration to measure its actual Type I error and power. That's just ONE candidate. To compare candidates and find a good one, you simulate each. And the OCs you measure depend on all the parameters in interaction, so you can't isolate one knob and tune it. This is what makes "complex designs" actually complex — not the underlying math but the lack of a single formula. That's the cost calibration removes, and it's why BATON exists. That sets up the pivot — the next slide.

The calibration challenge: four competing goals

Designing an adaptive trial requires achieving **all four** simultaneously.

Of thousands of plausible designs, only a small fraction satisfies all four — and you cannot find it by hand.

This is the slide that names what calibration actually has to do. The four goals come from the design problem itself, not from a methodologist's preference. (1) FDA compliance — the trial has to meet the regulatory floor on Type I error and power. Without this, no IND, no protocol. (2) Minimize patients — the smallest expected N consistent with the other constraints, because pancreatic cancer patients are a scarce, time-critical resource. (3) Stop bad early — when the drug isn't working, futility stopping protects patients from staying on an inactive arm. (4) Accelerate good — when the drug IS working, efficacy stopping gets the answer to the field faster. The visual point: all four at once. A design that hits FDA OCs but enrolls 200 patients fails goal 2. A design that minimizes N but doesn't meet power fails goal 1. A design that stops bad early but never declares efficacy fails goal 4. Calibration is finding the corner of parameter space where ALL four are satisfied simultaneously. Quantitatively, in our experience: of thousands of plausible parameter combinations, only a small fraction — typically 1-10% — meets all four. The exact fraction depends on the design and the OC targets; the symposium slide quotes "out of 1,000 possible designs, only ~70 satisfy all four requirements." Land that as the punchline: this is what BATON has to find. Pivot to next slide.

"We could design the science.

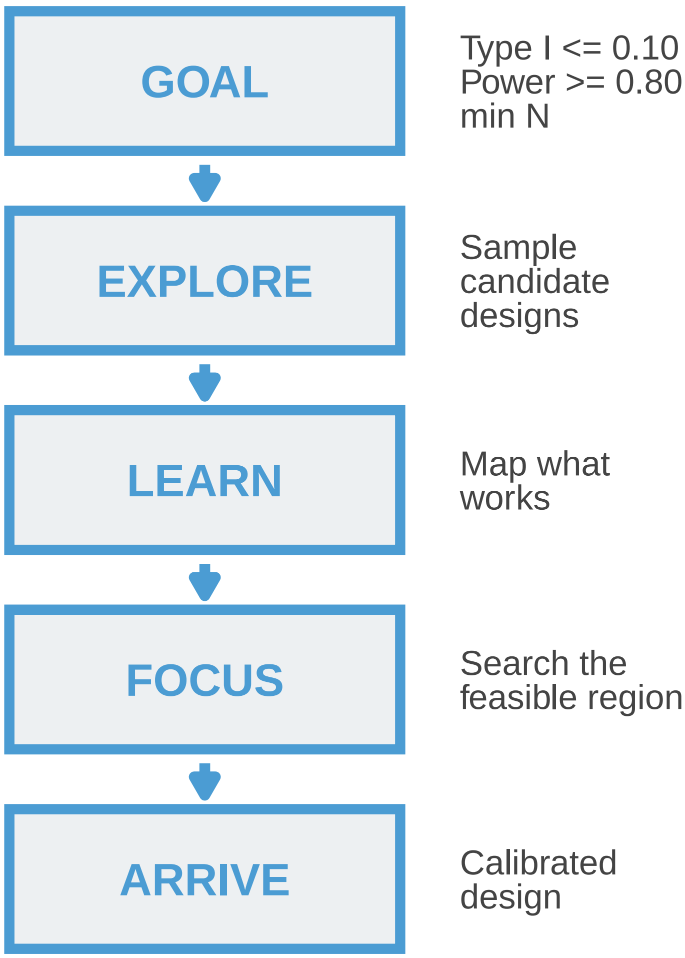
We could not design the trial."

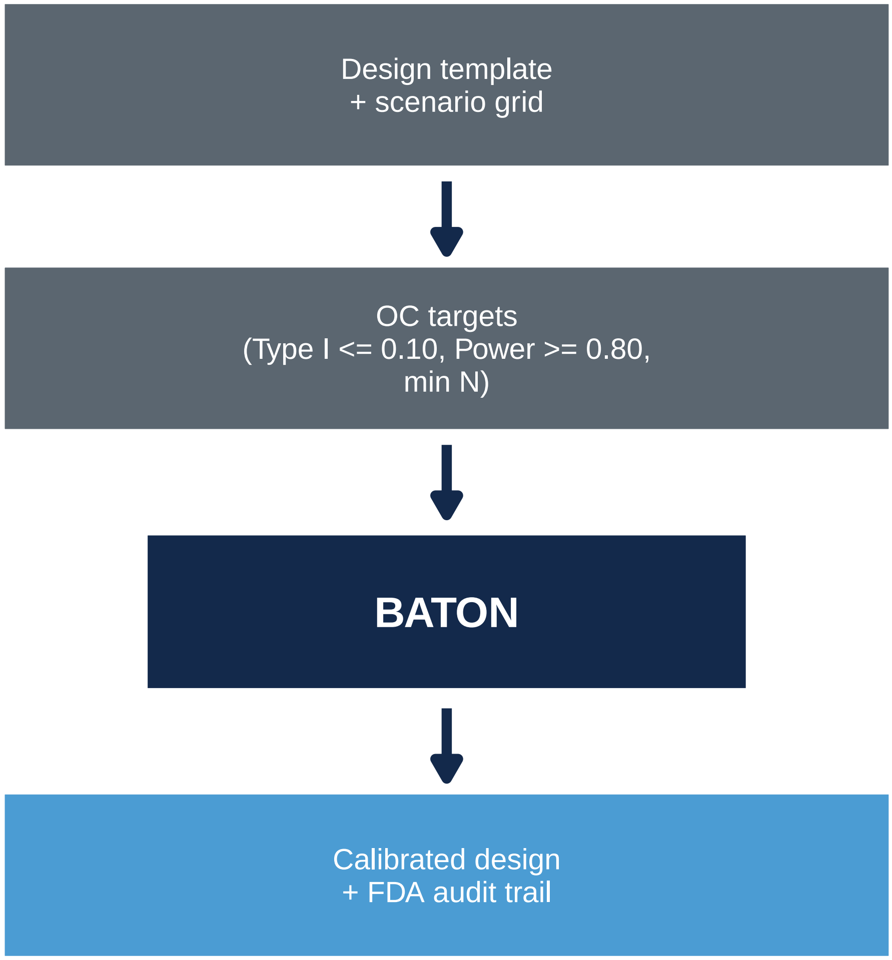
Pause. Let the line land. This is the pivot of the talk. Everything before this slide has been about establishing that the methods exist — combination dose-finding, OBD selection, resistance-aware longitudinal designs. ADAPT and EVOLVE prove these methods are deployable. But during EVOLVE's activation, the team kept running into the same wall: we knew what trial we wanted to run, we knew the statistical machinery existed, and we still could not get a calibrated, FDA-defensible design out the door. The methods existed. Deploying them did not. That is the gap. Pause briefly. Then move on to what closes it: BATON.

BATON: the calibration tool

BATON = **B**ayesian **A**daptive **T**rial **O**ptimizatio**N**[29]

BATON turns clinically necessary trial complexity into calibrated, launchable designs.



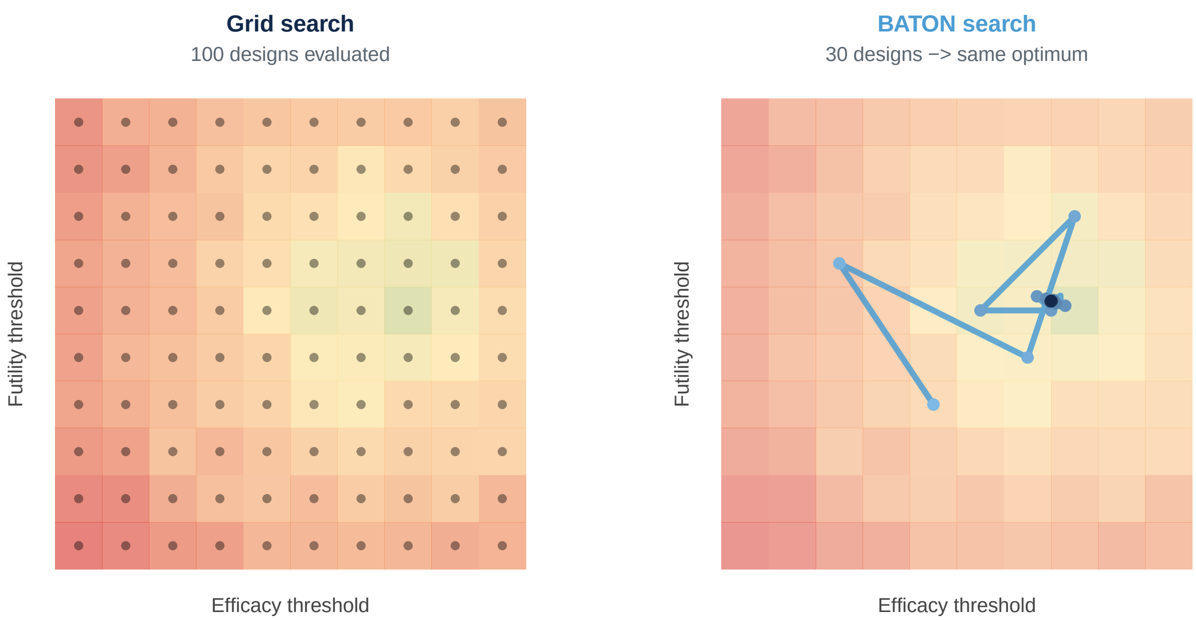


You bring the clinical inputs; BATON returns the calibration.

You bring the clinical design; BATON evaluates hundreds of candidates *in silico* and returns the ones that meet your targets. R package: github.com/naimurashid/BATON[29]

Think of BATON as GPS for trial design. You tell it the destination — power, type-I error, sample-size targets, expected duration. It explores candidate designs by Monte Carlo simulation. It learns which regions of the parameter space satisfy your constraints. It focuses the search on the feasible region. And it arrives at a calibrated design. Hours instead of months. Under the hood: constrained Bayesian optimization with Gaussian process surrogates, but for this audience what matters is the input and the output. You give BATON a design template, a set of operating-characteristic targets, and a scenario grid. BATON returns a calibrated design that meets your targets. FRAMING DISCIPLINE — and this matters because the next slide is PANGEA: in the BATON manuscript, the worked example is EVOLVE, not PANGEA. BATON was developed and validated during EVOLVE activation, across breast cancer subtrials at 8 to 13 parameters. PANGEA is an application of the same tool, not what the paper reports. If anyone asks "is this published," the clean answer is: the method and its validation are under review at JASA Applications and Case Studies; PANGEA is one application of the same software. Q&A POCKETS, in order of likelihood. (1) "Is BATON really optimal?" Manuscript benchmarked it against Simon two-stage designs where the optimum is known analytically; BATON converges to within 1% of Simon-optimal expected N, hitting the optimum exactly in 8 of 10 seeded runs. So the answer is: it recovers the analytically optimal design on problems where we can check. (2) "Can we use it?" Yes — open-source R package at github.com/naimurashid/BATON, with the companion evolveTrial package. (3) "What's the gain over grid or random search?" The paper has a head-to-head: BATON found feasible designs in 150–200 evaluations where grid and random search often found none at all. (4) "Why isn't a good statistician enough?" Honest answer: BATON IS the statistician's tool. A senior statistician with BATON > the same statistician without it. What BATON removes is the months-of-recalibration tax that currently makes the right designs operationally infeasible — the team can't commit to deliver a calibrated, defensible design within the trial timeline. Manual calibration also returns one design and you hope it's robust; BATON returns the feasible region and lets you pick the most defensible point in it. Under the hood for the methodologist who asks: constrained Bayesian optimization with Gaussian process surrogates over the operating-characteristic landscape.

Grid search vs BATON



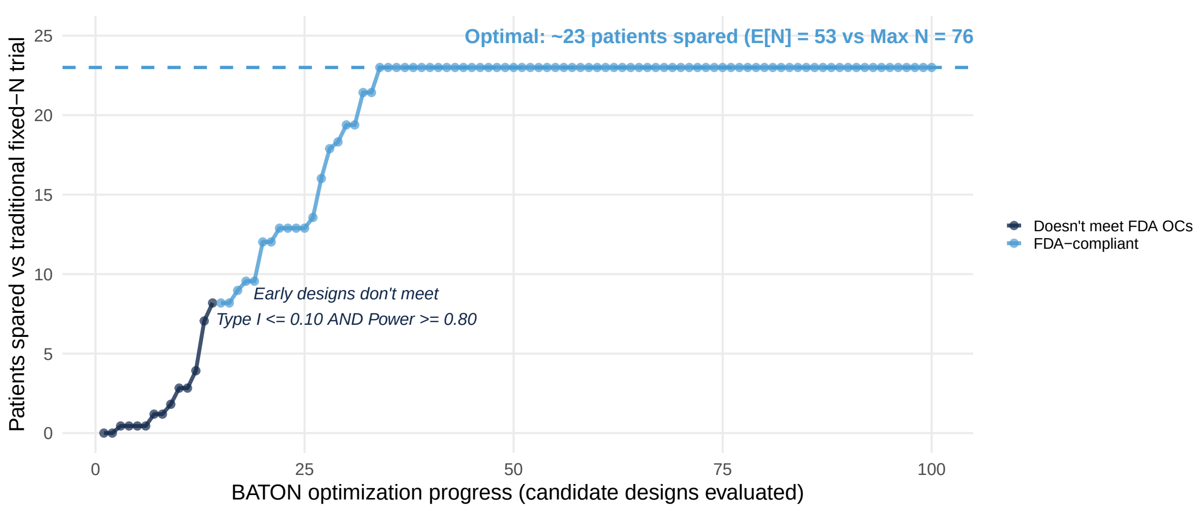
70% fewer evaluations — and the same optimum.

	Manual	BATON
Process	Trial-and-error	Systematic search
Wall-clock	Weeks to months	Hours
Output	One design, often under-validated	Full feasible region + FDA-grade audit trail

Same clinical question. Same statistical constraints. The bottleneck was the calibration.

The heatmap is the visual proof: grid search evaluates 100 designs spread across a 10x10 lattice; BATON evaluates 30 sequentially and converges on the same optimum because each evaluation informs the next. 70% fewer evaluations. Each evaluation is thousands of Monte Carlo trial replicates, so saving 70% of the search is the difference between hours and weeks. This is a 2D illustration; PANGEA Phase II's design space is 4-dimensional and EVOLVE designs go up to 8-13 dimensions, where the efficiency advantage compounds. The comparison row below tells the operational story: manual is trial-and-error with one design at a time, no reproducibility guarantee, often-incomplete documentation, variable regulatory risk; BATON is systematic, deterministic (seeded), hundreds evaluated, complete FDA audit trail. Same clinical question, same statistical constraints — the bottleneck was never the design itself, it was the calibration. The methodology shift Project Optimus formalized — adaptive Bayesian designs as the standard — is the destination. BATON is what makes getting there feasible at scale.

BATON's output, framed as patients spared



~23	7.0%	82.2%
patients spared (E[N] = 53 vs Max N = 76)	achieved Type I (target <= 10%)	achieved Power (target >= 80%)

Savings come from adaptive stopping; BATON's job is finding the design that achieves them within the OC constraints.

Each point on the convergence plot = one candidate design evaluated (10,000 simulated trials per design).

This is the same BATON convergence story you just saw on the grid-vs-BATON slide, but framed in clinically relevant units. Y-axis: patients spared relative to running a fixed-N=76 design. X-axis: BATON's optimization progress — each point is one candidate design that BATON evaluated by simulating ten thousand trials at that parameter combination. Dark-blue points are designs that don't yet meet the FDA operating-characteristic constraints — Type I error 0.10 AND power 0.80, simultaneously. Light-blue points are designs that do. BATON enters the feasible region around evaluation 14, then climbs to the optimum near evaluation 75. The horizontal dashed line — about 23 patients spared — is where the optimal calibrated design lands. CREDIT DISCIPLINE: the patients are spared by the adaptive design class itself — early futility/efficacy stopping is what makes E[N] of 53 possible against a max of 76. BATON's contribution is finding the configuration of stopping thresholds and futility margin that delivers those operating characteristics. Say "the BATON-calibrated adaptive design spares ~23 patients" or "the design BATON found spares ~23 patients", not "BATON spares ~23 patients" — the second sentence credits the optimizer rather than the design. The achieved expected N of 53 against the calibrated max of 76 is what you'll see on the next slide.

PANGEA Phase II: calibrated by BATON

- After Phase I OBD → **Phase II Bayesian adaptive RCT:** GnP + erlotinib (at OBD) vs GnP alone, primary PFS
- **Design assumes** reference 5.5 mo → 9.2 mo (HR 0.58)[30]
- **BATON-calibrated** four parameters under Type I 0.10, Power 0.80[29]

Achieved operating characteristics:

Metric	Value
Power	**82.2%**
Type I error	7.0%
Max N	**76** (82 accrual w/ buffer)
Expected N under H	53.0

Metric	Value
Expected N under H	58.1
Pr(early futility stop | H)	**77.9%**
Pr(early efficacy stop | H)	**81.4%**
Expected duration	27.3 mo

Without BATON: four interacting parameters rarely hit both constraints simultaneously. **With BATON:** joint calibration against both constraints, ~200 evaluations.

An unexpected finding: caps set too low suppress early-efficacy declarations. BATON returned the **Pareto frontier**; we chose 76 to preserve early-efficacy stopping. Manual calibration would have anchored on the first feasible design.

Von Hoff et al., *NEJM* 2013 (MPACT) · Young et al., 2026 (BATON manuscript)

Once U-BOIN selects the OBD, PANGEA transitions into a Phase II Bayesian adaptive randomized trial — basal-like PDAC patients randomized one-to-one to GnP plus erlotinib at the OBD versus GnP alone, with PFS as the primary endpoint. TENSE DISCIPLINE — critical, and the slide structure protects it: the 5.5-month reference, the 9.2-month alternative, and the HR 0.58 are DESIGN ASSUMPTIONS, not observed results. The 5.5 is the GnP doublet benchmark from MPACT, Von Hoff 2013. The trial is powered to detect HR 0.58, not showing one. The OCs in the right column ARE achieved, because they are outputs of BATON's calibration — Type I at 7.0%, power at 82.2%, computed over 10,000 simulated trials per scenario. Never let "BATON found a design that achieves these OCs" drift to "the trial showed this." Four design parameters had to be set jointly to meet OCs — max sample size, efficacy stopping threshold, futility stopping threshold, and futility hazard-ratio margin. CONFIRM AGAINST MANUSCRIPT which exact four BATON searched vs which were fixed by clinical/modeling choice. Three numbers to land: power 82% at Type I 7%; trial stops early for futility 78% of the time under H — patients are not randomized to a losing arm waiting for the readout; trial stops early for efficacy 81% of the time under H . Expected duration: 27.3 months; expected N: 53 under H , 58 under H , well below max 76. (Protocol carries 82 as accrual target with buffer; 76 is the calibrated max. If asked: "76 is calibrated max, 82 is accrual goal.") UNEXPECTED FINDING — the Pareto callout. We started with an instinct toward a tight max N to minimize regret if the drug doesn't work. BATON identified a non-obvious trade-off: pushing the cap too low actively SUPPRESSED early-efficacy declarations — patients wait until the final analysis for an answer even when the drug DOES work. The recommendation was a slightly higher cap (76 vs the tighter cap we were instinctively heading toward) that preserves both early-futility stopping AND early-efficacy stopping. Manual calibration would have anchored on the first feasible design it found. BATON returned the Pareto frontier; we chose where to land. This is the kind of insight manual calibration misses. WITHOUT/WITH BATON — quick read for the callout above. Without BATON: months of hand-tuning, four interacting parameters that rarely hit both constraints simultaneously, no reproducibility, no audit trail. With BATON: ~200 evaluations in hours, joint calibration against both constraints, deterministic seeded output, FDA-grade audit trail. The argument is not “the statistics got fancier”; the argument is “the bottleneck moved.” Calibration used to be the choke point. BATON moves it.

What this means for a PDAC patient

Mr. Hernandez, 62, mPDAC, basal-like by PurIST, progressed on 1L GnP.

Traditional fixed Phase II

- All **76 patients** enrolled before any look
- Drug effect discovered only at study end
- ~**36+ months** to read out

PANGEA Phase II (BATON-calibrated)

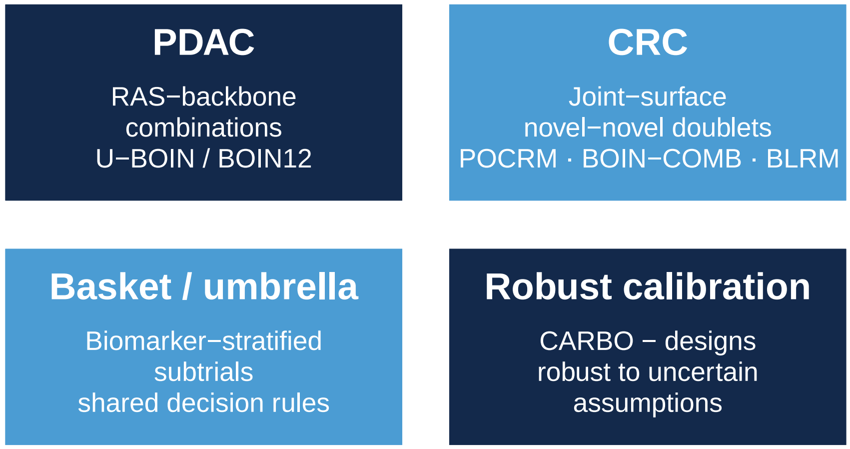
- **78% chance** of early futility stop
- E[N]: ~**53** (H) / ~**58** (H)
- **27.3 months** expected duration

Sooner answer, fewer patients on a losing arm — without losing power.

This is what calibrated PANGEA Phase II means for a real patient. Pick someone in your clinic. A 62-year-old with metastatic PDAC, basal-like by PurIST, progressed on first-line gemcitabine plus nab-paclitaxel. In a traditional fixed Phase II of GnP plus erlotinib versus GnP alone, that patient enrolls into a trial that will run all 76 patients before anyone learns whether erlotinib adds anything. Three-plus years to read out. In PANGEA Phase II, calibrated by BATON: same randomization, but interim analyses run as PFS events accumulate. The design has a seventy-eight percent chance of stopping early for futility if erlotinib adds nothing — patients are not randomized to an inactive arm beyond what’s needed to learn that. Expected enrollment is around 53 patients under H0 and 58 under H1. Expected duration: 27.3 months, not 36-plus. About 23 fewer patients exposed to a potentially-ineffective therapy. That is the clinical payoff of calibrating with BATON instead of shipping a fixed-N design that learns slowly. This is one trial. When the calibration becomes routine for the GI portfolio, the savings compound across every adaptive trial we activate.

Where this is going

BATON calibrates **aggressive**, **conservative**, or **balanced** designs — and ports across GI applications:

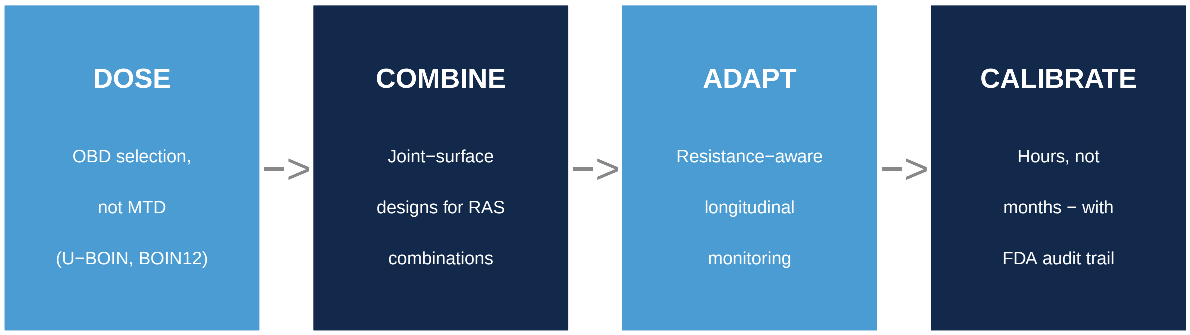


Could BATON become standard infrastructure for adaptive trial design in GI oncology?

Young et al., 2026 (BATON) · Wages 2011 · Lin & Yin 2017 · Neuenschwander 2008

Q&A POCKET — non-RAS examples of joint-surface combination dose-finding. If anyone asks "do any trials actually use joint-surface combination designs": yes, two clean concrete answers. (1) POCRM: the Embolden trial (NCT03240211) at UVA — pembrolizumab + pralatrexate/decitabine combination Phase Ib in T cell lymphomas, published implementation by Mozgunov et al, Stat Med 2022. (2) BLRM: Novartis Oncology has used BLRM as its standard Phase I design for over a decade and now exclusively for Phase 1 — including combination trials via the dual-agent extension. So joint-surface designs are not theoretical — they have published RCT deployments and one of the largest oncology developers in the world uses BLRM by default. The RAS-specific gap is that novel-novel RAS doublets like elironrasib + daraxonrasib are in the clinic but have not yet adopted these methods. That gap is exactly where BATON's joint-design extension lands. Where this is going, in two pieces. First, what's done. BATON has calibrated both EVOLVE — the single-arm, between-arm, and seamless two-stage breast cancer subtrials reported in the BATON manuscript — and PANGEA Phase II, the calibration you just saw. Two trials, two diseases, same tool. Second, the extensions in flight. The methodological move that matters most for this audience is bringing BATON's constrained-Bayesian-optimization machinery to the Phase I family — U-BOIN and BOIN12 — for OBD selection. PANGEA Phase I uses U-BOIN today, calibrated with the standard online software; the next step is doing that calibration under BATON so the multi-parameter Phase I design space stops being the bottleneck. That's how the calibration cost that has historically pushed PIs back toward 3+3 disappears. The second extension is CARBO — Context-Aware Robust Bayesian Optimization — which calibrates against a range of plausible trial assumptions rather than a single point estimate. This matters when the assumption was X but the trial actually enrolled at half the rate with 60% of the assumed biomarker prevalence; CARBO makes the design robust to that uncertainty. Mention CARBO in Q&A territory if time permits. The bigger point: when a modern Phase I/II design can be calibrated in days instead of months, the design choice becomes about what's right for the trial, not what's feasible in the timeline. The next generation of GI oncology trials should not be limited by what a statistician can hand-tune. We have the methods to do better, and the infrastructure to deploy them.

The new GI oncology trial template



Each piece is mature methodology. The gap was infrastructure — and we're closing it.

Visual summary of the talk in four boxes. DOSE — select on biology and benefit-risk, not toxicity alone. U-BOIN and BOIN12 deliver the OBD that Project Optimus is asking for. COMBINE — when you go from one drug to two, search the joint surface. POCRM, BOIN-COMB, BLRM, not sequential lead-ins. ADAPT — monitor for resistance during the trial, not after. ctDNA, serial biopsies, biomarker-stratified subtrials. CALIBRATE — BATON makes the parameter calibration tractable so the methods become deployable. Each piece is mature methodology. The gap was never the statistics; the gap was the infrastructure to deploy what we already had. We're closing it.

Take-home

1. **The RAS era needs new kinds of trials** — biomarker-guided, dose-optimized, resistance-aware.
2. **The methods exist.** Bayesian adaptive designs handle all three.
3. **BATON makes them practical to calibrate and launch.**

Three things to leave with. First, the RAS era needs a new generation of trials: biomarker-guided combinations, dose-optimized below the MTD where non-monotonic dose-response makes the MTD the wrong target anyway, and resistance-aware because patients fail therapy and we need to see it happen during the trial, not after. Second, the methods to run those trials exist. Bayesian adaptive designs are the tool family — they handle non-monotonic dose-response through utility-based OBD-finding, they handle combinations through joint admissibility on toxicity and efficacy, and they handle in-trial biomarker adaptation through pre-specified rules with simulation-calibrated error control. Third, the bottleneck has been making these designs practical to deploy. BATON is our answer to that. It calibrates the parameter settings against your operating-characteristic targets in days instead of months, so design choices become about what's right for the trial rather than what's feasible in the timeline. The thesis I opened with: new RAS therapies need new kinds of trials. Methods exist. The bottleneck is making those designs practical to calibrate and run. Thank you.

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Clinical collaborators

- Lisa Carey (UNC, EVOLVE PI)
- Channing Der (UNC, NEJM editorial[31])
- Jen Jen Yeh (UNC Surgery, SToP SPORE PI · NEJM editorial · PDAC collaborator)
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Questions

Thanks. Happy to take questions.